

Explicit uniform lower bounds for planar sets avoiding integral distances

A trimmed and vertically compressed digit construction

lixiang90*

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Abstract

We construct a single explicit unbounded open subset A of the Euclidean plane with no positive integral distance between distinct points and prove that, for every real $R \geq e$,

$$|A \cap B_R(0)| \geq \frac{9}{9604} \sqrt{R} \left(\frac{\log \log R}{\log R} \right)^3.$$

The construction trims the alphabet of a digit product, reduces its vertical scaling, and enlarges the surviving rectangles. Uniform estimates for entire rectangles establish compatibility both within and between infinitely many generations. A single rectangle of area $3/8$ suffices as the finite prefix. Compared with the reference construction used here, the certified coefficient improves by a factor $124416/2401$. Exact optimization within the specified parameter family has asymptotic improvement factor $384e^{-3/2}$. This is an explicit constant improvement with three logarithmic losses; neither global optimality of the coefficient nor a lower bound of order $\sqrt{R}$ is asserted.

1 Problem, scope, and main result

Write $B_R(0) = \{z \in \mathbb{R}^2 : |z| < R\}$ and let

$$M(R) = \sup\{|E| : E \subset B_R(0) \text{ is measurable, } |x - y| \notin \mathbb{Z}_{>0} \text{ } (x, y \in E, x \neq y)\}.$$

Here $|E|$ denotes two-dimensional Lebesgue measure and all logarithms are natural. Erdős Problem 953 asks for the growth of $M(R)$ [1]. Sárközy's digit construction supplies nearly square-root lower bounds for robustly avoiding integral distances [2, 3]; Goenka and Moore give a modern treatment and higher-dimensional extensions [4]. The square-root upper bound for the measurable problem is background to the present work [5]. We do not reprove or claim that upper bound.

Theorem 1.1. *There is an explicitly specified unbounded open set $A \subset \mathbb{R}^2$ such that no distance between distinct points of A is a positive integer, and*

$$|A \cap B_R(0)| \geq \frac{9}{9604} \sqrt{R} \left(\frac{\log \log R}{\log R} \right)^3 \quad (R \geq e). \quad (1)$$

Consequently $M(R)$ satisfies the same lower bound. The construction is fixed independently of R .

*Prepared with assistance from OpenAI Codex. The classical digit method is attributed to Sárközy. The existing upper-bound formalization in the companion repository is by Allen Hart and is not a contribution of this paper.

The coefficient in (1) is a guaranteed uniform coefficient, not a proved optimum for $M(R)$ or even the largest coefficient available from this set. The word “optimal” below refers exclusively to the finite choice of a trim parameter within the stated family. The exponent $1/2$ is unchanged, and the factor $(\log \log R / \log R)^3$ has not been removed.

2 Definition of the fixed infinite construction

For each integer $j \geq 6$, put

$$B_j = 2^j, \quad C_j = B_j^2/16,$$

let $n_j \geq 2$ be the largest integer with $(2n_j)^{2n_j} \leq C_j$, and set $k_j = \lfloor C_j^{1/(2n_j)} \rfloor$. Thus

$$2n_j \leq k_j, \quad k_j^{2n_j} \leq C_j < (k_j + 1)^{2n_j}, \quad C_j < [2(n_j + 1)]^{2(n_j + 1)}. \quad (2)$$

Existence follows because $C_j \geq 256$ and $(2n)^{2n}$ grows without bound. For $j \geq 14$ these parameters satisfy $k_j \geq 8$.

Temporarily write $k = k_j$, $n = n_j$. For $1 \leq t \leq \lfloor k/2 \rfloor$, define

$$q = k - t, \quad H(k, t) = \min \left\{ \frac{1}{256}, \frac{4t^2}{9k^3} \right\}. \quad (3)$$

For $\alpha = (\alpha_0, \dots, \alpha_{n-1}) \in \{0, \dots, q-1\}^n$, put

$$x_{j,\alpha} = B_j + \sum_{i < n} \alpha_i k^i, \quad (4)$$

$$y_{j,\alpha} = 3B_j^2 + k \sum_{i < n} \alpha_i k^{2i}. \quad (5)$$

Let $Q_{j,\alpha}$ be the open rectangle centred there, of full width $1/2$ and full height $H(k, t)$. Choose t_j to maximize

$$D_j(t) = \frac{1}{2}(k_j - t)^{n_j} H(k_j, t) \quad (6)$$

over its finite integer range, taking the smaller t in a tie. This rule is an exact finite comparison of rational numbers.

Define the base rectangle

$$P = (-3/8, 3/8) \times (-17/32, -1/32), \quad |P| = 3/8,$$

and the complete set

$$A = P \cup \bigcup_{j \geq 14} \bigcup_{\alpha \in \{0, \dots, k_j - t_j - 1\}^{n_j}} Q_{j,\alpha}. \quad (7)$$

This is a countable union of open sets, hence open and measurable. It is unbounded since even the zero digit string has centre $(2^j, 3 \cdot 4^j)$. Compared with the reference tail, the alphabet loses $t - 1$ symbols, the vertical multiplier changes from $8k$ to k , and the rectangles are enlarged. Thus (7) is not obtained merely by adding points to the old tail.

3 Entire rectangles within one generation

We give estimates independent of j, n and of any enumeration of pairs. Take two different strings, let r be their highest differing position, and orient their difference so that its leading digit is $d \geq 1$. Put

$$X = k^r, \quad a = |\Delta y|, \quad b = |\Delta x|, \quad p = \frac{q-1}{k-1}, \quad c = \frac{q-1}{k^2-1}.$$

Both coordinate differences have the sign of the leading digit, since the sum of all lower magnitudes is smaller than the leading place. In particular a, b are positive integers. Geometric sums give

$$(d - p)X \leq b \leq (d + p)X, \quad k(d - c)X^2 \leq a \leq k(d + c)X^2. \quad (8)$$

Here $p < 1$, $c < 1/(k + 1) \leq 1/9$, and $1 \leq d \leq q - 1 \leq k - 2$.

Lemma 3.1. *For all these pairs, with $H = H(k, t)$,*

$$(b - 1/2)^2 > 2aH.$$

Proof. If $r \geq 1$, retaining the exact geometric remainder gives

$$\begin{aligned} b &\geq dk^r - (k - t - 1) \sum_{i < r} k^i \\ &= (d - 1)k^r + t \sum_{i < r} k^i + 1 \geq dtk^{r-1} + 1. \end{aligned}$$

The last inequality uses $t \leq k$ and $\sum_{i < r} k^i \geq k^{r-1}$. With $L = tk^{r-1} > 0$, (8) and $H \leq 4t^2/(9k^3)$ give

$$2aH \leq \frac{8}{9}(d + 1/9)L^2.$$

Since $d \geq 1$,

$$d^2 - \frac{8}{9}(d + 1/9) \geq \frac{1}{81},$$

so $(b - 1/2)^2 > 2aH$. If $r = 0$, then $a = kd$, $b = d$ and $2t \leq k$; thus $2aH \leq 2d/9$ whereas $(d - 1/2)^2 - 2d/9 \geq d/36 > 0$. $\square$

Lemma 3.2. *For every such pair,*

$$(b + 1/2)^2 \leq \frac{45}{32}a.$$

Proof. Because $X \geq 1$, (8) implies $b + 1/2 \leq (d + 3/2)X$. The quadratic $(d + 3/2)^2 - (k + 2)d$ is convex in d on $[1, k - 2]$ and has equal endpoint values $17/4 - k < 0$. Therefore

$$(b + 1/2)^2 \leq (k + 2)dX^2.$$

Also $a \geq (8/9)kdX^2$, by $c \leq 1/9$ and $d \geq 1$. Since $k \geq 8$, $(k + 2)/k \leq 5/4$, proving the stated factor $45/32$. $\square$

For points in these two rectangles, horizontal and vertical differences can be written $b + u$, $a + v$ after taking absolute values, where $|u| \leq 1/2$, $|v| \leq H$. Lemma 3.1 shows

$$(b + u)^2 + (a + v)^2 \geq (b - 1/2)^2 + (a - H)^2 > a^2.$$

Lemma 3.2 and $H \leq 1/256$ show

$$\begin{aligned} (b + u)^2 + (a + v)^2 &\leq (b + 1/2)^2 + (a + H)^2 \\ &\leq a^2 + (45/32 + 1/128)a + H^2 < (a + 1)^2. \end{aligned}$$

Here $2 - (45/32 + 1/128) = 75/128 > 0$, $a \geq 1$ and $H^2 < 1$. All distances therefore lie in $(a, a + 1)$, which contains no integer. The diameter of one rectangle has square at most $1/4 + (1/256)^2 < 1$. Different horizontal centre coordinates are different integers, so the rectangles are disjoint, and their exact combined area is (6).

4 Compatibility across infinitely many generations

From geometric sums and (2), every centre in generation j lies in the frame

$$B_j \leq x < 5B_j/4, \quad 3B_j^2 \leq y < 7B_j^2/2. \quad (9)$$

Indeed the horizontal increment is less than $k^n \leq B_j/4$, and the vertical increment is less than $k^{2n} \leq B_j^2/16$; the displayed upper vertical bound is a convenient looser one.

Consider frames of two generations $i < j$, so $B_i \leq B_j/2$. If $a = y_j - y_i$ and $b = x_j - x_i$, then

$$b \geq 3B_j/8 \geq 48, \quad b^2 \leq a \leq 32b^2.$$

For example, $a \geq 17B_j^2/8$ and $b \leq 5B_j/4$ give the first square inequality; $a \leq 7B_j^2/2$ and $b \geq 3B_j/8$ give the second. Both a and b are integers.

Set $h = 1/256$ and $\delta = 1/512$. Two rectangles change b by at most $1/2$ and a by at most h . Consequently

$$\frac{95}{96}b \leq b + u \leq \frac{97}{96}b, \quad |v| \leq h.$$

The lower comparison with $(a + \delta)^2$ is strictly positive since

$$(95/96)^2 - 64(h + \delta) = 5569/9216 > 0, \quad h^2 - \delta^2 > 0.$$

The upper comparison with $(a + 1 - \delta)^2$ is strictly positive since

$$2(1 - h - \delta) - (97/96)^2 = 8915/9216 > 0, \quad (1 - \delta)^2 - h^2 > 0.$$

Thus all cross-generation distances lie in $(a + \delta, a + 1 - \delta)$. This proves compatibility of every future pair of generations, without truncating the infinite union.

For a centre (b, a) in any generation $j \geq 14$, (9) gives

$$b \geq 64, \quad a \leq 3b^2, \quad 48b^2 \leq 25a.$$

The base rectangle is contained in

$$C_* = \{(x, y) : x^2 + (y + 9/32)^2 < 1/4, \ -23/32 < y < 5/32\}.$$

Comparing a point of $P \subset C_*$ to a new rectangle, the horizontal perturbation is at most $3/4$ and the vertical perturbation belongs to $[-81/512, 369/512]$. Hence the horizontal square lies between $(253/256)^2 b^2$ and $(259/256)^2 b^2$. The coefficients for the same two square comparisons are respectively

$$(253/256)^2 - 6(81/512 + \delta) = 1033/65536 > 0,$$

$$2(1 - \delta - 369/512) - (25/48)(259/256)^2 = 67871/3145728 > 0.$$

The corresponding constant terms are positive. Distances from P to every new rectangle also lie in $(a + \delta, a + 1 - \delta)$. Finally the diameter of P has square $(3/4)^2 + (1/2)^2 = 13/16 < 1$. We have proved the distance-avoidance assertion of Theorem 1.1.

5 A uniform gain in the area of every tail generation

The reference area used to extract the previous uniform coefficient is

$$U_j = \frac{(k_j - 1)^{n_j}}{108k_j^3}.$$

It corresponds to full width $1/3$ and full height $1/(36k_j^3)$. For $k \geq 8$, the allowed choice $t = 2$ has uncapped height $16/(9k^3)$. Since our selected t_j maximizes (6),

$$\frac{D_j(t_j)}{U_j} \geq 96 \left(\frac{k_j - 2}{k_j - 1} \right)^{n_j} \geq G := 96(6/7)^4 = \frac{124416}{2401}. \quad (10)$$

For completeness, put $f(x) = (x/2) \log((x-2)/(x-1))$. Using $\log(1-u) \geq -u/(1-u)$ for $0 < u < 1$ gives, for $x \geq 8$,

$$f'(x) = \frac{1}{2} \log \frac{x-2}{x-1} + \frac{x}{2(x-2)(x-1)} \geq \frac{1}{2(x-2)(x-1)} > 0.$$

As $n \leq k/2$ and the base of the power is below one,

$$\left(\frac{k-2}{k-1} \right)^n \geq \exp(f(k)) \geq \exp(f(8)) = (6/7)^4.$$

This is a proof for every allowed parameter, not an extrapolation of computed examples. The first tail generation has $j = 14$, $(k, n, t) = (8, 4, 2)$ and exact area $D_{14} = 9/4$; its reference area is $2401/55296$.

6 Passage to every real radius

Let $R_j = 4^{j+1}$. The frame (9) and the rectangle dimensions imply that every rectangle of generation j lies strictly inside B_{R_j} . For example $x + y + 1 < 4B_j^2$ for $B_j \geq 64$ bounds the Euclidean norm. Every rectangle of generation $j+1$ lies outside B_{R_j} since its lower vertical coordinate exceeds $3 \cdot 4^{j+1} - 1/512 > R_j$. In particular

$$|A \cap B_{R_j}| = 3/8 + \sum_{14 \leq \ell \leq j} D_\ell \quad (j \geq 14). \quad (11)$$

Lemma 6.1. *Suppose a measurable distance-avoiding set E satisfies $|E \cap B_{R_j}| \geq GU_j$ for every $j \geq 6$, with $G > 0$. Then*

$$|E \cap B_R| \geq \frac{G}{55296} \sqrt{R} \left(\frac{\log \log R}{\log R} \right)^3 \quad (R \geq 16384).$$

Proof. Choose the unique $j \geq 6$ with $R_j \leq R < R_{j+1} = 4R_j$. Bernoulli's inequality, $2n \leq k$, and $C_j < (k+1)^{2n}$ give

$$(k-1)^n \geq k^n/2 \geq (k+1)^n/4 > \sqrt{C_j}/4 = B_j/16.$$

Therefore

$$U_j > \frac{B_j}{1728k_j^3} > \frac{\sqrt{R}}{6912k_j^3}. \quad (12)$$

For $n_j \geq 6$ we have the explicit base bound

$$k_j < \frac{2 \log R}{\log \log R}. \quad (13)$$

Here are details of the bound and its finite cases. Write $T = \log R$. The budget $2n \log(2n) < T$ implies $n < 3T/(4 \log T)$: for $s = 3T/(2 \log T)$, the tangent inequality for $\log$ gives $\log s > (2/3) \log T$, hence $s \log s > T$, and monotonicity of $x \log x$ completes the inversion. For $n \geq 16$, induction gives $2(n+1) \leq (5/4)^n$, beginning with $34 \cdot 4^{16} < 5^{16}$. Together with (2), this yields $k < (5/2)(n+1) \leq (8/3)n < 2T/\log T$. For $6 \leq n \leq 15$, the following integer maxima K bound k :

n	6	7	8	9	10	11	12	13	14	15
K	21	23	25	27	29	32	34	36	38	40

They follow by the exact inequality $[2(n+1)]^{2(n+1)} \leq (K+1)^{2n}$. Set $T_0(k) = \log 10 + 2n \log k$. The function $k \log T_0(k)/T_0(k)$ is increasing for $k \geq 2n$, while $\log T/T$ is decreasing for $T \geq T_0(k) > e$. For each displayed K , the inequality $K \log T_0(K) < 2T_0(K)$ is proved by rational logarithm bounds; these ten inequalities are also present as kernel-checked lemmas in the companion source. Their premises apply here because $R \geq 64C_j \geq 10k_j^{2n_j}$. Substituting (13) into (12) proves the lemma for $n_j \geq 6$.

The remaining generations are exactly $6 \leq j \leq 23$. They can be checked with the short table in Appendix A: throughout each whole interval $[R_j, R_{j+1})$, use $\sqrt{R} < 2^{j+2}$ and the indicated upper bound q_0 for $\log \log R / \log R$. Every table row satisfies the exact rational inequality

$$55296U_j > 2^{j+2}q_0^3.$$

These are interval checks and so cover real as well as integer radii. Finally area is monotone with radius, giving the claim. $\square$

For $j \geq 14$, (10) proves the premise of Lemma 6.1. For $6 \leq j \leq 13$, the base rectangle alone suffices: the exact values of GU_j are

j	6	7	8	9
GU_j	162/2401	18432/300125	9/196	115200/3195731
j	10	11	12	13
GU_j	2000/7203	9/28	104976/300125	2662/7203

Each is strictly below $3/8$. Thus for $R \geq 16384$ we get the coefficient $G/55296 = 9/9604$. For $e \leq R < 16384$, the entire base rectangle lies in B_R , while $\sqrt{R} \leq 128$ and $0 \leq \log \log R / \log R \leq 1$. Since $128 \cdot 9/9604 < 3/8$, this covers the remaining radii and completes the proof of Theorem 1.1.

The extremal quantity $M(R)$ independently satisfies $M(R) = \pi R^2$ for $0 < R \leq 1/2$ and $M(R) \geq \pi/4$ for $R \geq 1/2$, by open disks of radius at most $1/2$. Combining these with (1) gives a lower bound for $M(R)$ at every positive radius. The witnesses for those small-radius bounds need not coincide with the fixed infinite set A .

7 Optimization within the certified family

The guaranteed coefficient used only the admissible choice $t = 2$. Optimization generally does better. From (2),

$$2n \leq k < [2(n+1)]^{1+1/n}, \quad k/(2n) \longrightarrow 1.$$

In the uncapped region, maximizing (6) is equivalent to maximizing $t^2(k-t)^n$. Its logarithmic derivative is $2/t - n/(k-t)$, strictly decreasing, with unique zero

$$t_{\text{real}} = \frac{2k}{n+2} \longrightarrow 4.$$

The height cap begins at $t = \sqrt{9k^3/1024}$, which tends to infinity. Above that point the capped area decreases with t . Thus the cap does not affect the eventual maximizer. Comparison of the finitely many adjacent integers, or their limiting objective ratios, shows that the unique optimal integer is eventually $t = 4$. For clarity, the limiting objective is $t^2 e^{-t/2}$, whose unique real maximum is $t = 4$; its neighbouring integer values at 3 and 5 are strictly smaller, and log concavity excludes all other integers.

Consequently, for the reference per-generation areas,

$$\frac{D_j(t_j)}{U_j} = 24t_j^2 \left(\frac{k_j - t_j}{k_j - 1} \right)^{n_j} \rightarrow 384e^{-3/2} = 85.68198149 \dots$$

The fraction of surviving digit strings tends to $e^{-3/2}$, approximately 22.313%, while the area of each surviving rectangle is multiplied by 384. The same ratio holds for cumulative areas at completed-generation radii, since a finite prefix is negligible and the reference cumulative area diverges. No ratio limit at arbitrary partially completed radii is claimed.

This family cannot remove a logarithmic loss. The elementary parameter bound $k < 9n$ follows from (2): for $n \geq 2$, $[2(n+1)]^{1+1/n} < (9n)$ (one may use $2(n+1) \leq 3^n$ and $6(n+1) \leq 9n$). For any allowed t ,

$$\frac{D_j(t)}{U_j} \leq 24t^2 \left(1 - \frac{t-1}{k-1} \right)^n \leq 24t^2 e^{-(t-1)/9}.$$

The right side is bounded independently of j and t . Thus arbitrary choices of the trim parameter within this uniform-height digit product can improve the multiplier only by a bounded amount. Non-product digit selection or variable shapes would require a new argument.

8 Verification, reproducibility, and attribution

The companion repository is

<https://github.com/lixiang90/erdos953-lower-bound-lean>.

Its formal proofs use Lean 4.32.2 and Mathlib commit

905b95818eb32af7874a58b427f50c1711a5e96c.

The uniform inequality for the single fixed set, for every real $R \geq e$, has been checked in Lean as `Erdos953Lower.retreatInfinity_uniform_lower`. The combined existence statement, including openness, unboundedness, and exclusion of every positive integral distance, is `Erdos953Lower.fixed_unbounded_uniform_lower`. Its extremal-area consequence is `Erdos953Lower.retreat_lower_Mopen`. The supporting modules verify parameter existence, the finite maximizing trim rule and its guaranteed gain, the area of each generation, exclusion of all positive integral distances within and between generations, and compatibility with the base rectangle. The checked theorems use only Lean’s standard axioms `propext`, `Classical.choice`, and `Quot.sound`; there are no proof placeholders. The asymptotic maximizing trim $t = 4$ and the limiting improvement factor $384e^{-3/2}$ in Section 7 are proved analytically in this paper and are not part of the Lean verification claim. A rational certificate audit, finite pair enumeration, and a Lean theorem are distinct forms of evidence. Run `lake build Erdos953Retreat` to check this lower construction independently of the upper-bound project. The companion `scripts/build_paper.py` runs LaTeX until citations and cross-references are resolved and rejects unresolved-reference warnings.

The enlarged finite prefix from previous computations can be appended in place of P . It consists of the clipped disk C_* and 7,813 rational rectangles. Its compatibility with the new tail is certified by rational envelopes covering every future source frame. Those data give stronger finite-radius area values, but are unnecessary for the main construction (7) and its

coefficient. In particular the present analytic proof does not rely on enumerating those 7,813 rectangles.

The classical digit principle is due to Sárközy; our changes are the trimmed alphabet, compressed vertical scaling, whole-rectangle estimates, infinite compatible assembly, and explicit constant extraction. The separately authored Lean upper-bound proof by Allen Hart is fetched at commit `0d031f7212150df788f4fa38c26cfc3fc729f3d0` for the earlier growth-exponent theorem. It is neither used in this paper’s lower construction nor redistributed as our source. Award eligibility and a full extremal solution are not conclusions of this paper.

A Short exact checks for the initial scales

The following table contains the entire finite part of the complete-generation radius check. Here q_0 is an upper bound valid throughout the real interval $[4^{j+1}, 4^{j+2})$. All rows satisfy $512(k-1)^n > 2^{j+2}k^3q_0^3$, equivalently the inequality used in Lemma 6.1.

j	k	n	q_0	j	k	n	q_0	j	k	n	q_0
6	4	2	1/4	12	10	3	1/4	18	16	4	1/6
7	5	2	1/4	13	12	3	1/6	19	10	5	1/8
8	8	2	1/4	14	8	4	1/6	20	12	5	1/8
9	11	2	1/4	15	9	4	1/6	21	13	5	1/8
10	6	3	1/4	16	11	4	1/6	22	16	5	1/8
11	8	3	1/4	17	13	4	1/6	23	18	5	1/8

To verify the bounds q_0 , use

$$\log T \leq T/4 \ (T \geq 9), \quad \log T \leq T/6 \ (T \geq 18), \quad \log T \leq T/8 \ (T \geq 27).$$

They follow from the tangent inequality $\log T \leq \log T_0 + (T - T_0)/T_0$ at $T_0 = 9, 18, 27$, respectively, and the rationally verifiable bounds $\log 3 < 9/8$, $\log 18 < 3$. Also $\log 2 > 27/40$, so the respective first-generation log radii $14 \log 2$, $28 \log 2$, $40 \log 2$ exceed 9, 18, 27. For a positive rational x , scale to $x = 2^m y$, $1 \leq y \leq 2$, and use $z = (y - 1)/(y + 1)$ in

$$2 \sum_{r=0}^{s-1} \frac{z^{2r+1}}{2r+1} \leq \log y \leq 2 \sum_{r=0}^{s-1} \frac{z^{2r+1}}{2r+1} + \frac{2z^{2s+1}}{(2s+1)(1-z^2)}.$$

Every comparison then reduces to exact rational arithmetic. The companion Lean logarithm certificate uses a slightly looser remainder and six terms, which are sufficient for the constants displayed here.

References

- [1] T. F. Bloom, *Erdős Problem 953*, <https://www.erdosproblems.com/953>, accessed 4 October 2026.
- [2] A. Sárközy, *On distances near integers. I*, Studia Sci. Math. Hungar. **11** (1976), 37–50.
- [3] A. Sárközy, *On distances near integers. II*, Studia Sci. Math. Hungar. **11** (1976), 105–111.
- [4] R. Goenka and K. Moore, *Point sets avoiding near-integer distances*, arXiv:2605.06621v1 (2026), <https://arxiv.org/abs/2605.06621>.
- [5] *The order of growth of planar sets avoiding integer distances*, ULAM research preprint (2026), <https://www.ulam.ai/research/erdos953-short.pdf>.
- [6] A. Hart, *Erdős 953 formalization*, <https://github.com/AllenGrahamHart/FormalConjectures-Bench/tree/0d031f7212150df788f4fa38c26cfc3fc729f3d0/formalizations/erdos953>.